

Evan Jones
U.S. Olympic & Paralympic Committee Practicum
Bi-weekly progress report
Week ending February 9th

Mr. Schwartz,

Over the past two weeks since our last update quite a bit has come along in the project. After finalizing my project scope to solo Alpine skiing events, I spent a fair amount of time compiling important sport and event related information especially on ranking methods and regulation changes. This is where my sport specific data insight for the week lies. In 2004 the sport governing body the FIS made significant changes to majority of the alpine events in order to increase safety. Although adjustments for these regulation changes haven't been considered yet, I'm seeing pretty reasonable evidence that some of these safety measures such as reduced ski size and course design elements to reduce danger to the athletes have significantly impacted the longevity of athletes. This in turn has altered the distribution of athlete medaling by age in years since. One of my primary focuses over the coming weeks is researching the best ways to reconcile or handle these distributional changes (adjust vs. keeping the newer data only). In addition to these sports' specific insights a lot of progress was also made on the data and modeling side of things. From a data perspective, one key task was getting our data structured properly for input into our KDE model. This process involved a bit of cleaning, grouping, aggregation, and dropping of redundant or uninformative columns. During this process one key observation was the number of missing fields for athlete age. After some research I discovered the FIS keeps extremely up-to-date and accurate athlete bios which I could scrape from to fill out majority of our missing datapoints. Majority of my work the past week centered around building a python script that utilizes selenium to scrape athlete bios efficiently and safely from the FIS website and converts/reconciles this data with the existing database. Although time consuming, I believe this venture will result in more accurate models. Additionally, I believe that although this script in its present form is isolated to alpine skiing and the FIS website, on further research it appears that many other sports' governing bodies have similar systems in-place for their own websites. With this in mind, there is a potential that my selenium script could be slightly altered to be compatible with several other sports which would allow an efficient way to reconciling and validating USOPC internal databases with no cost public data. From a modeling perspective, I also constructed a preliminary test KDE model for Men's Super G. After some general hyperparameter tuning it quickly became clear that everything was working as expected and that this modeling approach is looking quite capable. In the following weeks extending this modeling to the other events and creating properly formatted country/athlete pipelines for use with the KDE models will be key priority. Included in the following pages is a schedule I created for the remaining weeks of my project. I believe this schedule should suffice for an overall project plan as well as the key items I plan to accomplish in the coming weeks. Let me know if there is anything missing or needs clarification, and I will be happy to provide it.

Best regards,

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Evan Jones: United States Olympic & Paralympic Committee Practicum Plan

Week	Week Ending	Planned	Completed	Other Notes
1	January 19 th	- Narrow sport scope to single discipline - Get familiarized with data structure and project goals - Exploratory data analysis with an eye towards data cleaning and formatting	- Alpine Skiing selected - Organized and learned about data structure - Identified droppable columns; Noticed fair amount of missing entries	
2	January 26 th	- Construct some histograms to test basic age to medaling frequency and relationship - Research Alpine 12 unique events; sport governing bodies; scoring and ranking mechanics - Apply some cursory data cleaning and formatting - Identify and correct datatypes for database import	- Created histogram with clear relationship - Extensive notes and annotations taken on specific sport and event details - Dropped columns which were mostly null or near duplicates - Fixed pandas typing issues - Create coding system for missing, blank, nan, and other edge case field entries - Researched methods for filling missing data	
3	February 2 nd	- Create local MySQL database and load in data - Create python Selenium script for datamining International Ski Federation website for athletes - Separate and split team events from solo events - Compare histograms for different gender/event pairs to check similarities (see if sports should be combined or not)	- Local database created and working - Selenium script created and working; filled most missing values for age; scraped profile photos for visualization purposes later. - Data separated by team and gender groupings - Utility of combining event data still unclear further analysis needed	

4	February 9 th	- Organize data into format ready for KDE - Apply KDE to Men's Super G as a test run; Tune bandwidth; Test target variable approach medaling in general vs. specific medal. - Compile list of regulation changes for events - Add documentation to Selenium script/notebook	- Data formatted for KDE; initial results look promising. - Created bandwidth tuning interactive sliders for ad-hoc experimentation and visualization - Compiled list of impactful regulation changes and possible data adjustment remedies - Selenium documentation added	
5	February 16 th	- Research masking methods for train/test given problem setup - Apply remedies for regulations and necessary data adjustments - Fit KDE models on all solo events - Begin work on midterm slide deck		
6	February 23 rd	- Create country historic and forward-looking pipeline profiles which at each Olympic date contain competitors used; unused; and medal outcomes - Create selenium script for updating country pipelines as new athletes are entered in the FIS database. - Finish midterm slide deck		Midterm slide deck/report due
7	March 2 nd	- Prototype a couple of supervised ML models in Sklearn taking as input a country's athlete pipeline and outputting a medal count estimation		
8	March 9 th	- Start using simulation with KDE results to form predictions of medaling based on age distribution and country pipeline		
9	March 16 th	- Compare supervised modeling results and KDE results. If both insufficient		

		look to other avenues of modeling - Explore features other than age; do they improve the model?		
10	March 23 rd	- Aggregate all production quality models and predictions - Detailed hyperparameter tuning - Pickle models for reuse		
11	March 30 th	- Begin work on making model prediction and ground truth results in a tableau friendly format - Start layout and design work on tableau dashboard structure		
12	April 6 th	- Finalize tableau dashboard - Organize and add documentation to repository		
13	April 13 th	- Prepare final report and touch up		
14	April 20 th	- Prepare final report and touch up		Final report due